



SY7T108 DLMS Firmware Release

RELEASE NOTES

June 2023

1 General Instructions

DLMS solution provides ready to use firmware solution on SY7T108G series, a single-phase electricity metering SOC that integrates 128 KB flash, 8K RAM and 4K SRAM. This package is intended to be compliant as per Indian standard IS-15959 with all the amendments. The user shall review this code and make necessary modifications before submitting it for certification testing.

The firmware was written using 'SY7T_M11_SDK_SHUNT_CT_V302' as the baseline. This SDK provides comprehensive documentation describing the SY7T108G families in details. For questions about hardware library and drivers, please review the demo meter package 'SY7T_M11_SDK_SHUNT_CT_V302'.

2 Device and Host Integration

The firmware binary code is a hex file available in the top folder (/bin) of this release. It can be loaded to SY7T108 ICs by any Arm Cortex M0 programming device, or Keil ARMCC's IDE with a ULINK-ME USB debugging device.

This firmware operates a demonstration electric power meter by controlling an SY7108G series IC (It will not operate with other silicon versions.)

The firmware configures UART2 as primary DLMS port and UART1 as secondary (electrical) port as per BIS requirements. UART0 is reserved for the command line interface (CLI). All UARTs are running at 9600 baud.

The firmware disables the command line interface through a macro (`#define CLI_ENABLE 0`) defined in `$/src/SY7T_M11/projects/demo_dlms/include/cli.h`. User can enable the CLI if needed through this macro. A command line interface is used for meter calibration and laboratory tests of the IC. It operates via a serial interface via the board's serial connector, configured at 9600 BAUD, 8N1, XON/XOFF. The serial commands are documented in the SY7T_M11_MPU_Parameters.pdf.

The firmware is configured for server physical address = 256 and logical address = 1. User should configure the client (Kalki explorer and Confirmation Test Tool (CTT)) accordingly.

3 Testing Coverage

This DLMS firmware is tested on the demo board version DB_AM11_0-1 and Kalki explorer is used as a DLMS client on Windows 10 PC. Please configure Kalki explorer with Physical address = 256 and logical address = 1. The following tests are done on both primary and secondary serial ports as per IS-15959 specifications:

- 4 bytes addressing and 1 byte addressing

- All 3 associations – public (client address=16, No password), Meter Reader (32, “Hello123”) and Utility (48, “SIL_SY7T_M11_pwd”).
- Instantaneous parameters
- Block load survey
- Billing parameters
- Daily load
- Tamperers
- Activity calendar
- Single action schedule
- All other DLMS objects
- EMC/EMI/Meter accuracy/RTC accuracy/Meter & RTC calibration as part of ‘SY7T_M11_Meter_SDK_V3.02’ package

4 Build Environment

This code was developed and built using Keil ARMCC in uVision v5.34. The compiler is configured to use ‘MicroLib’ and Cross-Module optimization enabled. The firmware size with the present build configuration is as follows:

Program Size: Code=73356 RO-data=18384 RW-data=1144 ZI-data=5288

Note that, with Cross-Module optimisation enabled it takes 3 iterations for the build whereas with Cross-Module optimisation disabled, the build takes only one iteration, but the program size increases by 5 Kbytes. Also note that, additional 25 Kbytes of flash size is needed if CLI is compiled in.

5 Changes or Revision History

June 2023 – Version 3.04 - SY7T108_DLMS_V0304

- Features modifications as below-
 - Present + 12 billing history from existing present + 6.
 - Time zones reconfigured to 5 from existing 4.
 - Seasons reconfigured to 2 from existing 3.
 - Added over voltage tamper condition. Low Voltage tamper condition is configured in the application as phase voltage between 60 volts to 110 volts.
 - Added average current and active energy maximum demand to the load survey history.
 - Increased load survey history to 60 days from present 45 days.
 - Replaced self-diagnostics object – 1.0.96.5.0.255. This object is writable from the US association.
 - This build uses 128 Kbytes EEPROM. Refer to SY7T108_DLMS_EEPROM_map_v2.xlsx for the EEPROM address map used in the source code.

Mar 2023 – Version 3.03 - SY7T108_DLMS_V0303

- Reconfigured serial ports. UART0 is for CLI if enabled, UART2 is primary DLMS port and UART1 is secondary DLMS port.

June 2020 – Version 3.02 - SY7T_M11_DLMS_V0302

- DLMS solution for SY7T108G

April 2020 – Version 3.00 - SY7T_M11_DLMS_V0300_alpha

- First release of DLMS solution for SY7T108G.

6 Known Issues or Cautions

- Time Zones accumulations in the metering solution is for 5 time zones (time_of_day). For example – If TODs are 0-5, 5-10, 10-15 and 15-20 & 20-MidNight, then TZ0 = 0-5, TZ1 = 5-10, TZ2 = 10-15, TZ3 = 15-20 and TZ4 = 20-MidNight.
- 'Script Selector' in single action schedule object is writable but DLMS application does not process the 'Script Selector' (Time Zone accumulations are as per TOD only).
- Make sure, VAUX jumper (P37) is connected between 1&2 on demo board version labelled DB_AM11_0-1.PcbDoc for DLMS UART ports to work.